

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that the browser attempted to send a DNS request to the DNS server using UDP. This is based on the network analysis, which shows that the ICMP response returned the error message “udp port 53 unreachable.” Port 53 is used for DNS services, which translate domain names into IP addresses. The most likely issue is that the DNS service is unavailable, not listening on UDP port 53, or otherwise unreachable.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred at 1:24 PM when customers reported that they could not access yummyrecipesforme.com and received a “destination port unreachable” error. The cybersecurity team began investigating the issue and conducted packet-sniffing tests using tcpdump. The resulting log file showed that DNS port 53 was unreachable. The next step is to determine whether the DNS server is down or whether traffic to port 53 is being blocked by a firewall. One possible cause is that the DNS server is unavailable due to a misconfiguration or a successful Denial-of-Service attack.